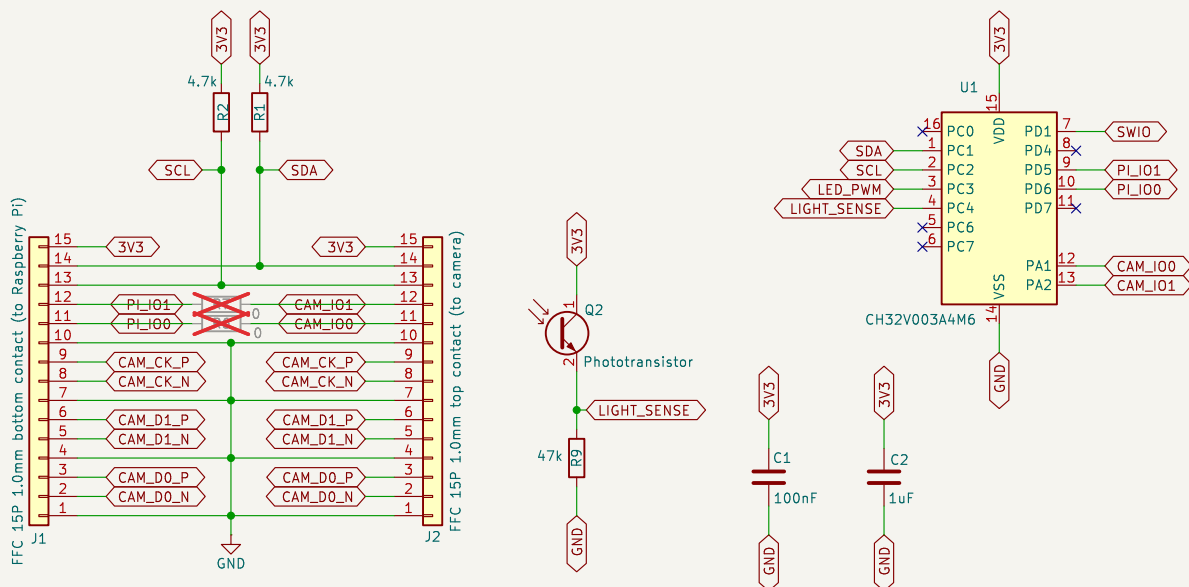
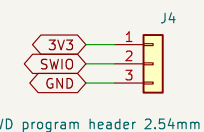
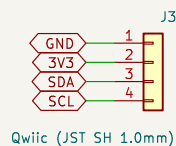


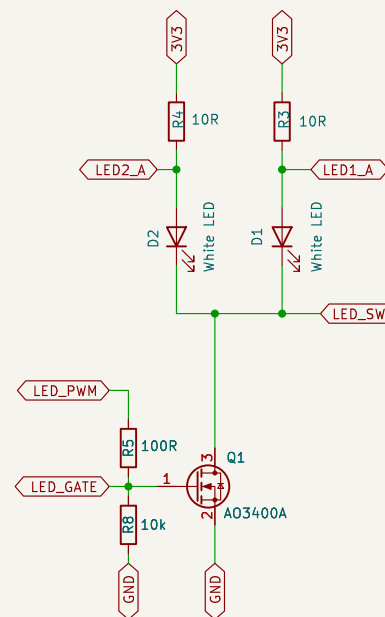
RPi Camera LED interposer: sits between a Raspberry Pi and its camera.
 MIPI lanes, I2C, GND and 3V3 pass straight through J1 -> J2.
 U1 (I2C slave on the camera bus) drives the camera GPIO pins 11/12,
 PWM-drives the illumination LEDs and reads ambient light on its ADC.



R6/R7 (OR, DNP) are a bypass: fit them to wire the Pi's CAM_IO pins straight to the camera, e.g. when U1 is unpopulated. Left off, U1 sits in the middle -- it watches the PI on PD5/PD6 and drives the camera on PA1/PA2. Do not fit them while U1 drives: OR gives contention no current limit.



LED current is set by R3/R4 against the 3V3 rail (about 30-40mA each with $V_f=2.9V$). Total draw comes from the Pi camera 3V3 supply -- keep the duty/current within the Pi's camera connector budget.



J1 and J2 are deliberately different parts: J1 bottom contact, J2 top contact. Top contact mirrors J2's cable-to-pad order, which cancels the 180 deg rotation between two connectors whose cables leave opposite edges -- so every J1->J2 net runs straight across on one layer instead of crossing the whole bus. Cost: the camera-side cable goes in contacts up, i.e. flipped from the Pi side. Use an opposite-side (type D) FFC there, or fold a standard same-side one.

SOP-16 bonds no two ports to one pin, so PD1/SWIO is programming only and PC3/T1CH3 drives the LED gate: no LED flicker while flashing firmware. PC6, PC7, PD4 and PC0 are spare. PD7 is left unused on purpose: it is NRST until the RST_MODE option byte says otherwise, so a Pi GPIO on it would reset the MCU on a factory-fresh part.

All parts chosen for JLCPCB / NextPCB Rev0 assembly; see research/*.md for availability and pricing (checked 2026-09-01).

CH32V003A4M6 camera GPIO / illumination / light sensor

Sheet: /
 File: rpi-camera-led.kicad_sch

Title: RPi Camera LED interposer

Size: A4 Date: 2026-09-01

KiCad E.D.A. 9.0.7

Rev: A

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